EMMANUEL MULILO

~Computer Science Student

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Summary

I prefer hands-on execution and figuring things out over waiting for the right moment. Driven by a deep curiosity for how complex systems operate and how they're built.

Education

The Copperbelt University Kitwe, Copperbelt

BSc. Computer Science, Undergraduate Nov, 2023 - Present

Relevant courses

- Database Systems: Relational Models, EER Diagrams, Normalization of Database Tables, SQL.
- Software Engineering: Software Processes, System Modeling, Project Management.
- Data Structures, Algorithms & Programming in C++: Complexity, Linked Lists, Recursion & Advanced Sorting and Searching Algorithms.
- Digital Design: Digital Logic Gates.
- Computer Architecture and Organization: Boolean Algebra, Performance Metrics & Processor Design.
- Operating Systems and Resource Management: File Systems and Management ,Memory Management, IPC & Protection.

Technical Skills

- Languages & Core: C++, SQL (MySQL Workbench)
- Operating Systems: Linux, Windows, OpenWrt (Embedded Firmware)
- AI & Agentic Workflows: Local RAG Pipelines, Ollama, OpenCode, Model Context Protocol (MCP), Vector Search / Embeddings
- Networking & Systems: Multi-Router Bridging, Traffic Shaping & QoS (SQM / CAKE), Local Network Configuration, Hardware Troubleshooting
- Developer Tools & Productivity: Git, Github, Office Suite (Word/Docs, Excel/Sheets, PowerPoint/Slides)

Projects

1. **Autonomous Portfolio Management Agent**
    - Implemented ReAct Agentic Workflow
    - Managed Scoped File Permissions
    - Optimized Model Token Routing
    - Integrated Git Version Control
2. **Air-Gapped Notes: A Privacy-First Local AI Research & Editing Assistant**
    - Engineered 100% Offline RAG Pipeline
    - Resource-Optimized Edge Computing
    - Implemented Semantic Knowledge Retrieval
3. **SmartQueue LTE: Low-Latency Network Optimization Gateway**
    - **Engineered a multi-tier dual-router network architecture** (bridged an LTE modem into Layer 2 passthrough mode and configured a secondary router).
    - **Mitigated severe cellular bufferbloat** by Smart Queue Management (SQM) via the CAKE algorithm and `cake-autorate`.
    - **Optimized physical and kernel-level performance** by resolving local subnets, managing link-layer framing overhead.